

1. Determine the network and broadcast addresses and number of host bits and hosts for the given IPv4 addresses and prefixes:

IPv4 Address/Prefix	Network Address	Broadcast Address	Total Number of Host Bits	Total Number of Hosts
172.16.104.99/27	172.16.16.96	172.16.16.127	5	$2^5 - 2 = 30$
198.133.219.250/24	198.133.219.0	198.133.219.255	8	$2^8 - 2$
10.1.113.75/19	10.1.96.0	10.1.127.255	13	2^{13}

IP: 172.16.104.99/27 = 10101100.00010000.01101000.01100011

NM: /27 (primii 27 de biti sunt 1) = 11111111.11111111.11111111.11100000

IP and NM:

10101100.0001000001101000.01100011

11111111.11111111.11111111.11100000 AND

10101100.0001000001101000.01100000 – Network Address (bitii de host sunt 0) 172.16.16.96

10101100.0001000001101000.01111111 – Broadcast Address (bitii de host sunt 1) 172.16.16.127

IP: 198.133.219.250/24 = 11000110.10000101.11011011.11111010

NM: /24 (primii 24 de biti sunt 1) = 11111111.11111111.11111111.00000000

IP and NM:

11000110.10000101.11011011.11111010

11111111.11111111.11111111.00000000 AND

11000110.10000101.11011011.00000000 – Network Address 198.133.219.0

11000110.10000101.11011011.11111111 – Broadcast Address 198.133.219.255

IP: 10.1.113.75/19 = 00001010.00000001.01110001.01001011

NM: /19 (primii 19 de biti sunt 1) = 11111111.11111111.11100000.00000000

IP and NM:

00001010.00000001.01110001.01001011

11111111.11111111.11100000.00000000

_____AND

00001010.00000001.01100000.00000000 – Network Address 10.1.96.0

00001010.00000001.01111111.11111111 – Broadcast Address 10.1.127.255

2. Having the following information, compute subnets with the following constraints:

- A number of 62 subnets

Host IP Address: 172.16.0.0

Original Subnet Mask 255.255.0.0

=> 6 biti pentru 62 subnets.

NM: 11111111.11111111.11111100.00000000

255.255.252.0

10 biti ramasi pentru hosts/ subnet

3. Having the following information, compute subnets with the following constraints:

- A maximum number of 29 hosts/subnet

Host IP Address: 192.168.200.0

Original Subnet Mask 255.255.255.0

⇒ Din 5 biti => 29 hosts putem avea (2 la puterea a 5 a e 32)

3 biti pentru subnet => 8 subnets

4. Having the following information, compute subnets with the following constraints:

- A number of 250 subnets

Host IP Address: 10.0.0.0

Original Subnet Mask 255.0.0.0

2 la a 8 a =256 => 8 biti pentru 250 subnets

NM: 11111111.11111111.00000000.00000000

255.255.0.0

=>16 biti host